

Immunohistochemistry (IHC) for In-situ Hybridization (ISH)-stained Mouse Sections

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Intended for 50-80 micron free-floating mouse tissue sections that were ISH-stained with BM Purple and post-fixed in 4%. All washing steps should be performed with gentle agitation on an orbital shaker.

Day 1

- 1. Remove the 4% PFA and wash sections for 2 X 5 min at 25°C in TBS containing 0.1% TritonX-100 (TBSTx).
- 2. Incubate sections for 5 min at 25°C in 3.0% H₂O₂ in 1X TBSTx.
- 3. Incubate sections for 25 min at 25°C in 0.3% H₂O₂ in 1X TBSTx.
- 4. Wash sections 4 X 5 min at 25°C in TBSTx.
- 5. Incubate sections for 20 min at 25°C in avidin blocking solution (Vector Labs# SP-2001).
- 6. Rinse sections briefly with TBSTx.
- 7. Incubate sections for 20 min at 25°C in biotin blocking solution (Vector Labs# SP-2001).
- 8. Wash sections 2 X 4 min at 25°C in TBSTx.
- 9. For IHC with primary antibodies made in mouse species (for non-mouse primary antibodies, skip to step #10):
 - A. Incubate sections for 1 hr at 25°C in a working solution of M.O.M. Mouse IgG Blocking Reagent (Vector Labs #MKB-2213).
 - B. Wash sections for 2 X 3 min at 25°C in TBSTx.
 - C. Proceed to step #11.
- 10. For IHC with primary antibodies made in non-mouse species:
- Incubate sections for 1 hr at 25°C in TBSTx containing 1% bovine serum albumin (Fisher #BP1600-100), 5% heat-inactivated serum from host-animal species of secondary antibody, and 1% Blocking Reagent (Roche Applied Science # 11096176001, diluted from a 10% stock solution of Blocking Reagent dissolved in maleic acid buffer as directed by the manufacturer). This Blocking Buffer is referred to as RBTx buffer.

- **11.** Incubate sections overnight at 4°C in RBTx buffer containing primary antibody (concentration determined empirically).

DAY 2

- **12.** Remove primary antibody and wash sections for 6 x 5 min at 25°C in TBSTx.
- **13.** Incubate sections for 1 hr at 25°C in RBTx containing biotinylated goat anti-mouse, goat anti-rabbit, or rabbit anti-goat secondary antibodies (Vector Lab # BA-9200, BA-1000, BA-5000, concentration determined empirically).
- **14.** Towards the end of step #13, dilute avidin biotin complex (ABC) reagent (Vector Labs # PK6100) in TBSTx according to the manufacturer's instructions and incubate for at least 30 min at 25°C.
- **15.** Wash sections for 6 x 5 min at 25°C in TBSTx.
- **16.** Incubate sections 45 min at 25°C in diluted ABC reagent.
- **17.** Wash 6 x 5 min at 25°C in TBSTx.
- **18.** Prepare 3, 3'-diaminobenzidine (DAB) solution (Vector Labs # SK-4100) in TBSTx as described by the manufacturer.
- **19.** Remove sections from tube and place in petridish. Remove TBSTx and add 1X DAB solution onto sections and monitor under microscope for color development (appr. 1-10 min).
- **20.** As soon as color becomes visible, dilute the DAB with excess TBSTw and post-fix in 4% PFA.